Electric field of Point Particles

Questions, Textbook; Answers, Elias Xu

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Two charged particles are attached to an x axis: Particle 1 of charge $-2.00 \times 10^{-7}C$ is at position $6.00cm$ and particle 2 of charge $2 \times 10^{-7}C$ is at position $x = 21.0cm$. Midway between the particles, what is their net electric field in unit-vector notation.

Given the fact that the particle in midway, the electical fields point in the same direction. The resulting calculation is simple:

$$d_{mid} = \frac{0.21\text{m} + 0.06\text{m}}{2} = 0.075\text{m}$$

$$E_{net} = E_1(r) - E_2(r) = \frac{1}{4\pi\epsilon_0}\frac{-2 \times 10^{-7}\text{C}}{(0.075\text{m})^2} - \frac{1}{4\pi\epsilon_0}\frac{2 \times 10^{-7}\text{C}}{(0.075\text{m})^2}$$

$$= \frac{1}{4\pi\epsilon_0(0.075\text{m})^2}\left(-2 \times 10^{-7}\text{C} - 2 \times 10^{-7}\text{C}\right) = \boxed{-6.392 \times 10^5 \frac{\text{N}}{\text{C}}}$$

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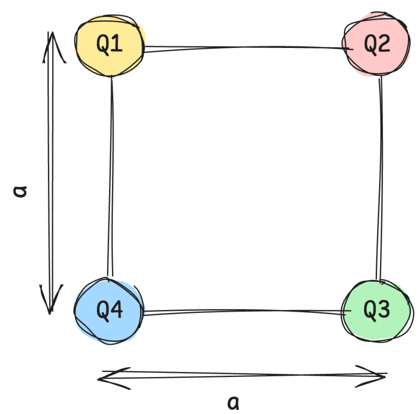
In the figure below, the four particles form a square of edge length $a = 5.00cm$ and have charges $q_1 = +10nC$; $q_2 = -20nC$; $q_3 = -20nC$; $q_4 = -10nC$. In unit-vector notation, what net electric field do the paricles produce at the square's center?



To solve this problem, one has to split into x and y components:

$$D = \sqrt{(0.025\text{m})^2 + (0.025\text{m})^2}$$

$$E_x = \frac{1}{4\pi\epsilon_0 D^2}\cos 45\,(10\text{nC} - 20\text{nC} + 20\text{nC} - 10\text{nC}) = 0$$

$$E_y = \frac{1}{4\pi\epsilon_0 D^2}\sin 45\,(-10\text{nC} + 20\text{nC} + 20\text{nC} - 10\text{nC}) = \frac{1}{4\pi\epsilon_0 D^2}\sin(45)20 \times 10^{-9}C = \boxed{101730.65\frac{\text{N}}{\text{C}}}$$